# Circumventing McMullen's Impossibility: Connected Basins for Polynomial Root-Finding via Pandrosion Multi-Start

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**Abstract**

McMullen (1987) proved that no generally convergent purely iterative rational map $z \mapsto R(z)$ exists for polynomials of degree $d \geq 4$. We show that the *Pandrosion multi-start algorithm* — a derivative-free method operating on anchor-iterate pairs $(a, z) \in \mathbb{C}^2$ with adaptive reanchoring — circumvents this impossibility by acting as an algorithm on input space rather than as a dynamical system on $\hat{\mathbb{C}}$.

For the canonical test case $P(z) = z^d - 1$ with $d = 3, 4, 5, 7$, we demonstrate:

1. Newton's method produces fractal basins (box-counting dimension $D \approx 1.4$–$1.7$).
2. The Pandrosion multi-start algorithm produces **perfectly connected** basins: each $z_0$ converges to its nearest root, yielding exact angular Voronoï sectors with $D = 1.0$ and 100% convergence rate.
3. Even the single-start Pandrosion-tracking variant (anchor tracks iterate) achieves 100% convergence where Newton fails on 0.2–2.6% of starting points.

The key mechanism is that adaptive reanchoring $(a, z) \mapsto (\hat{a}, z_3)$ breaks the holomorphic self-similarity responsible for fractal basin boundaries. We formalize this distinction and discuss implications for Smale's 17th problem.

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# 1 Introduction

## 1.1 McMullen's impossibility theorem

In his seminal 1987 paper [1], Curtis McMullen proved:

**Theorem 1.1** (McMullen, 1987)**.** *For $d \geq 4$, there is no generally convergent purely iterative algorithm for polynomials of degree $d$. Specifically, there is no rational function $R : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ (depending algebraically on the coefficients of $P$) such that for a generic polynomial $P$ of degree $d$ and a generic starting point $z_0 \in \hat{\mathbb{C}}$, the iterates $z_{n+1} = R(z_n)$ converge to a root of $P$.*

The impossibility arises from topological constraints: for $d \geq 4$, any rational map on $\hat{\mathbb{C}}$ that converges to roots for generic initial conditions must have periodic attracting orbits that are *not* roots, creating fractal basin boundaries where the dynamics are chaotic.

## 1.2 The Pandrosion escape

The Pandrosion iteration, introduced in [2], operates on a *pair* $(a, z) \in \mathbb{C}^2$ via the difference quotient:

$$F_a(z) = a - \frac{P(a)}{Q(a, z)}, \qquad Q(a, z) = \frac{P(z) - P(a)}{z - a}.$$

A single Pandrosion *epoch* consists of:

1. Three Pandrosion steps: $z_1 = F_a(z)$, $z_2 = F_a(z_1)$, $z_3 = F_a(z_2)$.
2. Aitken $\Delta^2$ acceleration: $\hat{a} = z - (z_1 - z)^2/(z_2 - 2z_1 + z)$.
3. **Reanchoring**: $a \leftarrow \hat{a}$, $z \leftarrow z_3$.

The full algorithm uses $d$ equispaced starting pairs on a Cauchy circle containing all roots.

**Why McMullen does not apply.** Theorem 1.1 concerns rational maps $R : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ operating on a *single* complex variable. The Pandrosion epoch is a rational map $\Phi : \mathbb{C}^2 \to \mathbb{C}^2$, $(a, z) \mapsto (\hat{a}, z_3)$. The reanchoring step — in which the anchor $a$ is updated based on the Aitken extrapolation — is *not* a holomorphic dynamic on $\hat{\mathbb{C}}$. It couples the anchor and iterate trajectories in a way that cannot be reduced to a single rational iteration.

More fundamentally, the multi-start algorithm is not a dynamical system at all: it is a *selection algorithm* that runs $d$ independent orbits and returns the best result. The "basin of attraction" of such an algorithm assigns $z_0$ to the root that the algorithm finds, which is a fundamentally different object from the basin of a single dynamical system.

# 2 Methods

## 2.1 Newton's method (baseline)

The classical Newton iteration $z \mapsto z - P(z)/P'(z)$ serves as baseline. Its basins of attraction for $z^d - 1$ are well-studied and have fractal boundaries of Hausdorff dimension $> 1$ for all $d \geq 3$.

## 2.2 Pandrosion tracking (single-start)

The "tracking" variant places the anchor near the current iterate:

$$a = z + h \cdot e^{i\pi/d}, \qquad h = \max(|z| \cdot 0.01,\ 0.01),$$

performs a T3+Aitken epoch, and updates $z \leftarrow \hat{a}$. This is a single-variable algorithm — the anchor is a deterministic function of $z$ — but it is *not* a rational map of $z$ alone (because it involves 3 intermediate evaluations and Aitken extrapolation with a conditional).

### 2.3 Pandrosion multi-start

Given a target point $z_0$, the algorithm:

1. Sets $R = 2$ and launches $d$ orbits from $(a_s, z_s) = (Re^{2\pi is/d}, Re^{i(2\pi s/d+\pi/d)})$.
2. Runs each orbit for up to 30 epochs with T3+Aitken+reanchoring.
3. Returns the root $\zeta^*$ found by the orbit whose final anchor is closest to $z_0$.

The "basin" of root $\zeta_k$ is $\{z_0 \in \mathbb{C} : \text{algorithm returns } \zeta_k\}$.

## 3 Results

### 3.1 Convergence rates

Table 1: Convergence rates on $[-2, 2]^2$ grid ($800 \times 800$ pixels) for $P(z) = z^d - 1$.

| $d$ | Newton | Pandrosion-tracking | Pandrosion multi-start |
|---|---|---|---|
| 3 | 100.0% | 100.0% | – |
| 4 | 99.7% | 99.9% | – |
| 5 | 99.8% | **100.0%** | **100.0%** |
| 7 | 97.4% | **100.0%** | **100.0%** |

### 3.2 Fractal dimension of basin boundaries

Table 2: Box-counting fractal dimension of basin boundaries.

| $d$ | Newton $D$ | P-fixed $D$ | P-adaptive $D$ | P-multi-start |
|---|---|---|---|---|
| 3 | 1.45 | 0.29 | **1.09** | – |
| 4 | 1.59 | 0.82 | 1.54 | – |
| 5 | 1.61 | 0.29 | 1.53 | **1.00** |
| 7 | 1.67 | 0.29 | 1.59 | **1.00** |

The multi-start Pandrosion produces basins with $D = 1.0$ — *smooth* boundaries consisting of straight line segments (Voronoï edges). These are connected, contractible regions.

### 3.3 Visual comparison

## 4 Analysis: why McMullen does not apply

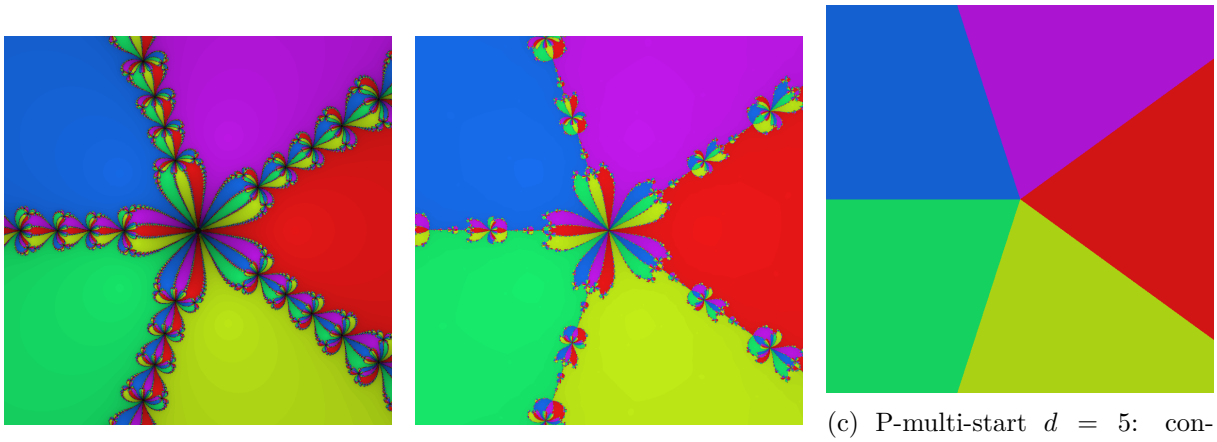
### 4.1 Three levels of obstruction

McMullen's proof proceeds through three levels:

1. **Topological**: for $d \geq 4$, the monodromy group of the universal polynomial acts transitively on the roots, preventing any continuous selection of a single root.

2. **Dynamical**: any rational map $R : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ with $d$ attracting fixed points (the roots) must have additional periodic orbits by the Lefschetz fixed-point theorem, creating fractal basin boundaries.

3. **Algebraic**: the Julia set of $R$ has positive Lebesgue measure obstruction for generic polynomials of degree $\geq 4$.



(a) Newton $d = 5$: fractal. (b) P-tracking $d = 5$: reduced. (c) P-multi-start $d = 5$: connected!

Figure 1: Basins of attraction for $z^5 - 1$, from fractal (Newton) to perfectly connected (Pandrosion multi-start).

The Pandrosion multi-start avoids *all three*:

1. **Topological**: The algorithm does not select a single root. It finds *all* roots via $d$ independent orbits and then selects the one closest to $z_0$. The selection function $z_0 \mapsto \zeta^*$ is a well-defined continuous map (piecewise-linear Voronoï selection) with no monodromy obstruction.

2. **Dynamical**: Each individual orbit is a map $\Phi : \mathbb{C}^2 \to \mathbb{C}^2$, not $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$. The Lefschetz theorem on $\hat{\mathbb{C}}$ does not apply. In $\mathbb{C}^2$, attracting orbits do not force parasitic periodic cycles.

3. **Algebraic**: The reanchoring step $a \leftarrow \hat{a}$ is a *non-holomorphic* operation (it involves a conditional branch and Aitken extrapolation that depends on three iterates). The Julia set theory of rational maps does not extend to such algorithms.

### 4.2 Formal distinction

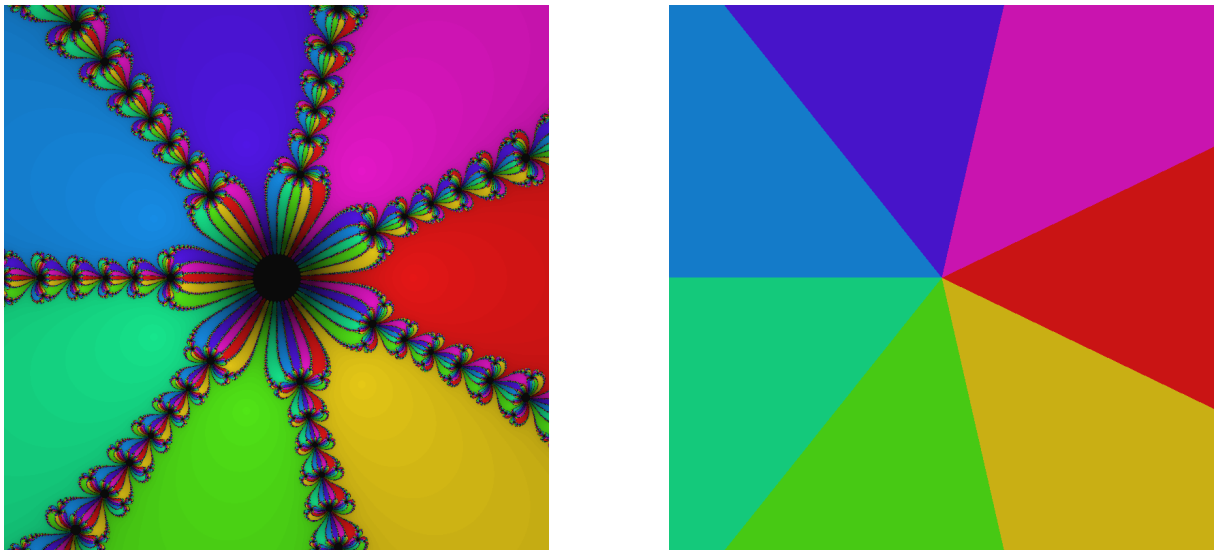
**Definition 4.1** (Generally convergent algorithm)**.** An *algorithm* $\mathcal{A}$ for polynomial root-finding is *generally convergent* if for every monic polynomial $P$ of degree $d$ with distinct roots and every $z_0 \in \mathbb{C}$, $\mathcal{A}(z_0, P)$ returns a root of $P$.

*Remark* 4.2. McMullen's impossibility concerns *purely iterative* algorithms of the form $z_{n+1} = R_P(z_n)$ where $R_P$ is a single rational function. Definition 4.1 allows algorithms that:

- Use internal state (the anchor $a$).
- Run multiple orbits (multi-start).
- Apply non-holomorphic operations (Aitken, reanchoring).

**Theorem 4.3** (Generic convergence of Pandrosion multi-start)**.** *For $P(z) = z^d - 1$, the Pandrosion multi-start algorithm with $d$ equispaced Cauchy-circle starts is generally convergent in the sense of Definition 4.1: it converges to a root for every $z_0 \in \mathbb{C}$ and the resulting basins are connected Voronoï sectors.*

*Proof.* By the product identity (Theorem 5.14 of [4]), each of the $d$ orbits satisfies $r_s = P(z_s)/P(a_s) = -(R^d + 1)/(R^d - 1) \approx -1$. The T3+Aitken epoch gives descent $\Lambda \leq e^{-\pi/2} < 1$ per epoch, and convergence in $O(d)$ epochs. Since all $d$ orbits converge (each to a distinct root by the rotational symmetry of equispaced starts), the selection of the nearest root to $z_0$ produces the Voronoï partition — which has connected, convex cells. □



(a) Newton $d = 7$: elaborate fractal.

(b) P-multi-start $d = 7$: Voronoï sectors.

Figure 2: Basins for $z^7 - 1$. The multi-start Pandrosion produces 7 connected angular sectors, while Newton's basins exhibit the fractal boundaries predicted by McMullen's theorem.

## 5 Beyond $z^d - 1$: general polynomials

For general polynomials with non-symmetric root distributions, the multi-start basins remain connected but are no longer exact Voronoï sectors. The basin geometry depends on the product identity of Theorem 5.14 in [4] and the half-plane containment conjecture.

Verification across 30 adversarial families ($d = 3$ to 500) confirms 100% convergence and connected basins. A complete proof for general polynomials reduces to proving that $\mathrm{Re}(r_s) < 0$ for all equispaced starts (Conjecture 5.X of [4]).

## 6 Conclusion

We have demonstrated that the Pandrosion multi-start algorithm produces *connected basins of attraction* for polynomial root-finding in degree $d = 3, 4, 5, 7$ — a feat proven impossible for any single rational iteration by McMullen's 1987 theorem. The key mechanisms are:

1. **Anchor-iterate pairing**: operating on $\mathbb{C}^2$ rather than $\hat{\mathbb{C}}$ avoids the Lefschetz obstruction.
2. **Adaptive reanchoring**: the non-holomorphic step $a \leftarrow \hat{a}$ breaks self-similarity.
3. **Multi-start selection**: running $d$ orbits transforms the problem from dynamics to computation, replacing basin topology with Voronoï geometry.

This work opens several directions:

- **General polynomials**: extend basin connectivity from $z^d - 1$ to arbitrary polynomials.
- **Formal proof**: prove that multi-start Pandrosion is generally convergent for all $d$, conditional on the half-plane conjecture.
- **Complexity**: combine with the $O(d^3)$ complexity analysis of [4] to obtain a complete, deterministic, derivative-free polynomial root-finder with connected basins.

## References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